

Phase 5: Project Development Phase

Development Report

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Location of Source Code

For compilation and execution safety, the runnable source code files are not duplicated inside this documentation folder. They are placed directly in the repository's root directory under three distinct sub-directories:

- **Dataset/:** Contains the raw data.
- **Training/:** Contains the Jupyter Notebook for data science tasks.
- **Flask/:** Contains the web server and serialized models.

Source Code Overview

- **Training/model_training.ipynb:** This script handles data loading from the Dataset folder, Label Encoding, Exploratory Data Analysis (using Seaborn), model comparison, RandomizedSearchCV hyperparameter tuning, 5-fold cross-validation, and exporting the final tuned model via `pickle.dump()`.
- **Flask/app.py:** This is the deployment script. It loads the serialized `.pkl` files, maps URL routes (`/`, `/apply`, `/predict`), handles string-to-integer conversion for incoming web form data, enforces column ordering, and renders HTML templates.

Dataset Path

The dataset utilized for training and testing is located at: `Dataset/loan_prediction.csv`